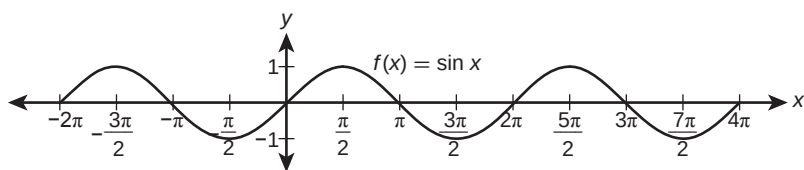


Sinusoidal Graphs: Period

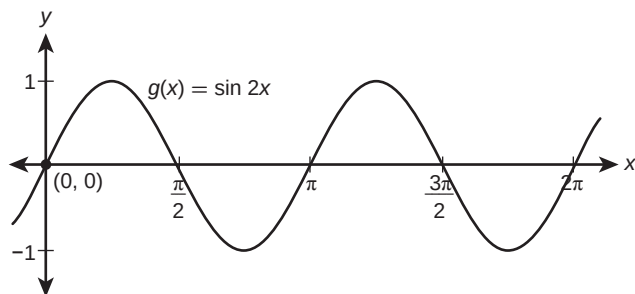
The interval over which a sinusoidal graph repeats itself is its period.

Determining the Period of Sine Functions

Recall that the period of a function is the smallest interval over which the function repeats itself. For the parent sine function, $f(x) = \sin x$, the period is 2π .



Compare the graph of the parent function $f(x)$ to the graph of $g(x) = \sin 2x$.



Notice that the period for $g(x) = \sin 2x$ is π . The graph has the same basic shape as $f(x) = \sin x$ but has been compressed horizontally by a factor of $\frac{1}{2}$.

Formula for the Period of Sine

For $f(x) = \sin Bx$, period $= \frac{2\pi}{B}$ where $B > 0$.

The function $f(x) = \sin 2x$ is in the form $f(x) = \sin Bx$, so its period is

$$\frac{2\pi}{B} = \frac{2\pi}{2} = \pi, \text{ which is one-half the period of the sine function.}$$

EXAMPLE 1

Use the formula for period to determine the period of

$g(x) = \sin\left(\frac{x}{3}\right)$. Then graph the function to check your answer.

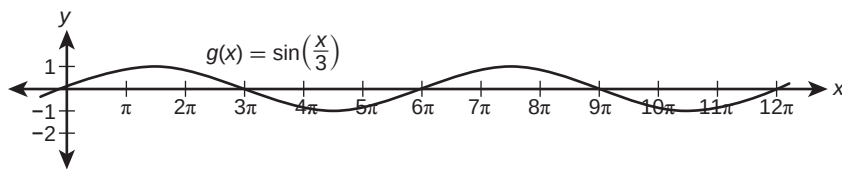
► **Think About It** Another way to define period is as the smallest real number p that makes $f(x) = f(x + p)$.

SOLUTION

Rewrite the function in the form $g(x) = \sin Bx$.

$$g(x) = \sin\left(\frac{x}{3}\right) = \sin\left(\frac{1}{3}x\right)$$

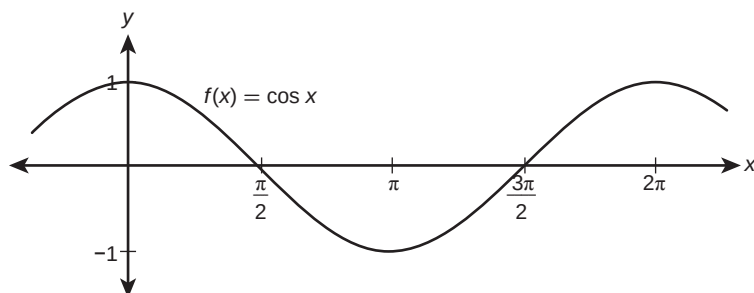
Since $B = \frac{1}{3}$, period $= \frac{2\pi}{\frac{1}{3}} = 6\pi$.



The graph also repeats every 6π , so the period is 6π . ■

Determining the Period of Cosine Functions

Recall that the period of the cosine function, $f(x) = \cos x$, is also 2π .



As with sine functions, the period of cosine functions can be found using a formula.

Formula for the Period of Cosine

For $f(x) = \cos Bx$, period = $\frac{2\pi}{B}$ where $B > 0$.

EXAMPLE 2

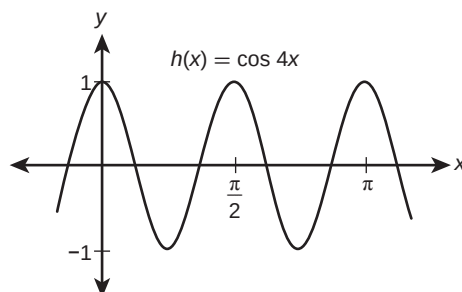
Use the formula for period to determine the period of $h(x) = \cos 4x$. Then graph the function to check your answer.

SOLUTION

The function is in the form $h(x) = \cos Bx$.

Since $B = 4$, period = $\frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$.

The graph repeats every $\frac{\pi}{2}$, so the period is $\frac{\pi}{2}$.



EXAMPLE 3

Determine the period of $f(x) = \cos \pi x$.

SOLUTION

Since $B = \pi$, period $= \frac{2\pi}{\pi} = 2$. ■

Determining Frequency

Definition

The **frequency** of a sinusoidal function is the number of periods per unit. A function's frequency is the reciprocal of its period.

Formula for Frequency

For $f(x) = \sin Bx$ and $f(x) = \cos Bx$, frequency $= \frac{1}{\text{period}} = \frac{B}{2\pi}$ where $B > 0$.

EXAMPLE 4

Determine the frequency of $g(x) = \sin 4x$.

SOLUTION

Determine the period. Since $B = 4$, period $= \frac{2\pi}{4} = \frac{\pi}{2}$. Take the reciprocal of the period.

$$\text{frequency} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi} \quad \blacksquare$$

EXAMPLE 5

Write the equation of a function in the form $f(x) = \cos Bx$ that has a frequency of $\frac{2}{3}$.

SOLUTION

$$\text{frequency} = \frac{2}{3} = \frac{B}{2\pi}$$

Solve for B .

$$4\pi = 3B$$

$$\frac{4\pi}{3} = B$$

An equation of the function is $f(x) = \cos\left(\frac{4\pi x}{3}\right)$. ■